

Project One-Pager

AI Chatbot for Social Engineering Attack Detection

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1. Actual Idea (Problem & Need)

Social Engineering attacks (phishing, smishing, fake OTP messages, urgent account warnings) are one of the most common ways attackers trick users into revealing personal or financial information. Many people fail to identify these dangerous messages because attackers use emotional triggers, urgency, and clever wording.

Idea:

Build an AI-powered **Chatbot** that takes a user's message and instantly tells whether it is a **Safe** message or a **Social Engineering Attack (Unsafe)** using machine learning and NLP.

This chatbot acts as a **defense tool**, helping users recognize suspicious communication.

2. Project Summary

This is a **full-stack web application** that detects social engineering attacks in real time.

- The **Frontend** is developed in **Next.js**, providing a chat interface where the user types a message.
- The **Backend** is developed using **FastAPI**, which receives the message and passes it through an ML model.
- The **Machine Learning Layer** uses **TF-IDF vectorization** and a **Logistic Regression model** trained on the SMS Spam Collection Dataset.
- The model returns **Safe** or **Unsafe**, allowing users to see whether the message resembles known social engineering attack patterns.

The chatbot is fast, responsive, intelligent, and capable of detecting threats instantly.

3. What Has Been Done & Working

Frontend (Next.js)

- Fully working chat interface
- Shows “Safe” or “Unsafe” labels in real-time
- Clean UI/UX
- Option to download chat history as `.txt`

Backend (FastAPI)

- `/predict` API endpoint working
- Loads pre-trained ML model and vectorizer
- Processes messages and sends prediction to frontend
- High performance using Uvicorn server

Machine Learning (Scikit-learn)

- Dataset cleaned and preprocessed
- TF-IDF vectorizer trained
- Logistic Regression model trained and saved (`.pkl`)
- Prediction accuracy good enough for live classification
- Additional rule-based checks for improved detection:
 - URLs
 - OTP keywords

- “Urgent”, “Account locked”, “Click here” patterns

Overall System

Chatbot running end-to-end

Real-time prediction working

Input → Backend → ML Model → Output flow verified

Project working locally without errors

4. Conclusion

The project successfully demonstrates how **AI and NLP** can be used to combat **Social Engineering Attacks**. By combining machine learning with a user-friendly chat interface, the system provides an effective way for people to quickly check suspicious messages.

This solution can be extended further to integrate with email systems, SMS filtering services, or cybersecurity awareness platforms.

Overall, the project meets its objective of creating a practical, intelligent, and efficient tool to help users stay safe from cyber threats.